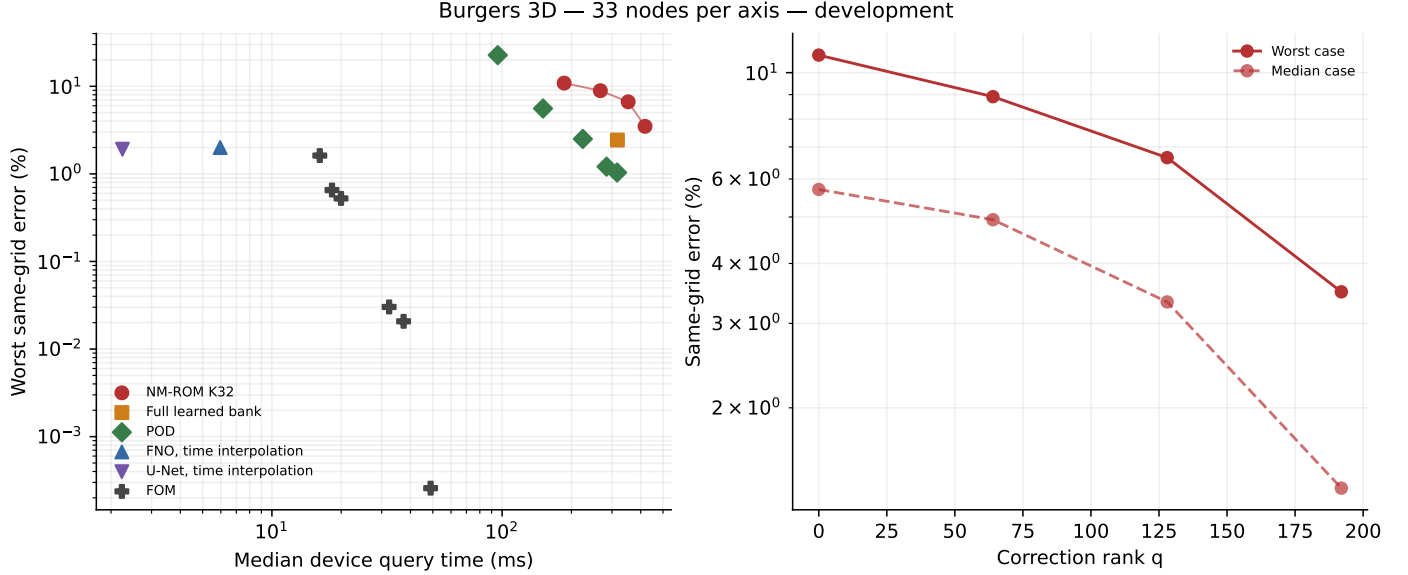


3D comparison panels — provisional development results

All values are generated from retained, audited invocation records. Final evaluation remains pending.

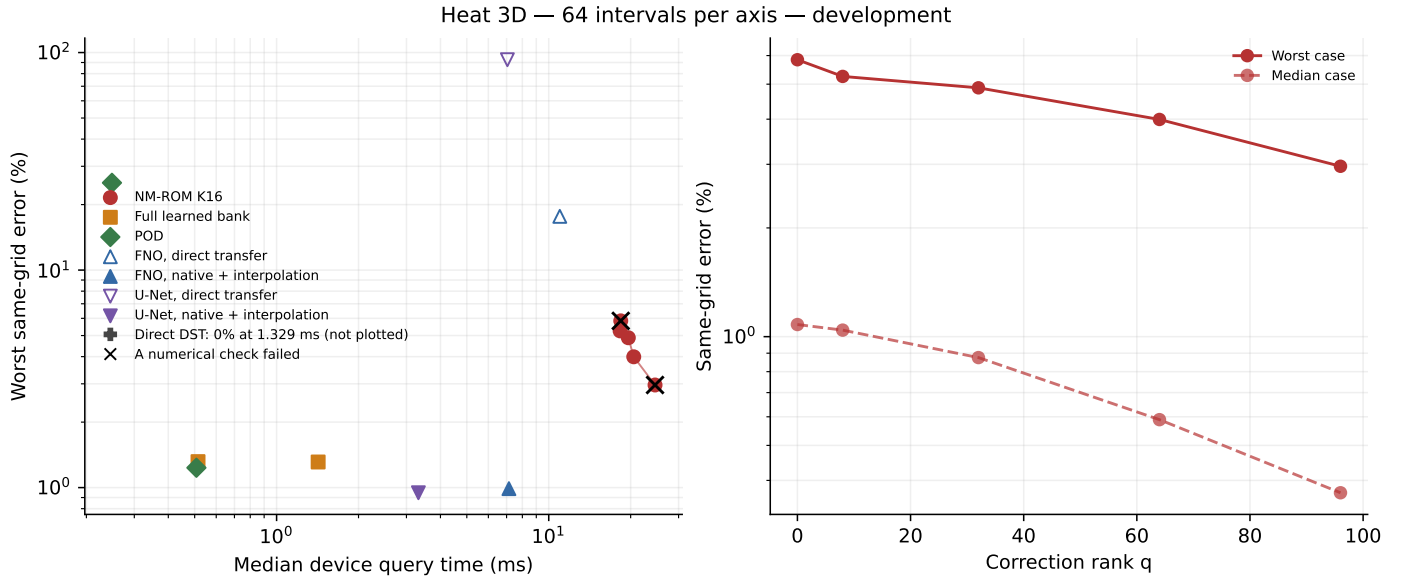
Method	Median (%)	Worst (%)	GPU ms	p95 ms	NF/NS	Outliers
NM-ROM K32, q0	5.7071	10.8780	185.964	230.676	0/0	0/24
NM-ROM K32, q64	4.9359	8.9108	267.228	324.793	0/0	0/24
NM-ROM K32, q128	3.3258	6.6491	352.971	427.954	0/0	0/24
NM-ROM K32, q192	1.3600	3.4908	417.752	619.396	0/0	0/24
Full learned bank R256	0.8437	2.4345	316.894	317.808	0/0	0/24
POD32	9.5224	22.6947	95.765	96.092	0/0	0/24
POD96	1.9855	5.5785	150.552	151.571	0/0	0/24
POD160	0.7675	2.5057	224.232	224.922	0/0	0/24
POD224	0.3988	1.2100	284.363	285.417	0/0	0/24
POD256	0.3292	1.0363	315.911	317.242	0/0	0/24
FNO, time interpolation	1.6413	1.9979	5.959	6.426	0/0	0/24
U-Net, time interpolation	1.6239	1.9129	2.242	2.508	0/0	0/24
FOM n1e-02, 15e-01	1.2739	1.6185	16.141	16.597	0/0	0/24
FOM n1e-02, 15e-03	0.0382	0.6543	18.239	21.011	0/0	0/24
FOM n1e-03, 15e-01	0.4692	0.5248	19.954	21.066	0/0	0/24
FOM n1e-04, 15e-01	0.0267	0.0303	32.302	34.894	0/0	0/24
FOM n1e-04, 11e-06	0.0188	0.0207	37.326	41.866	0/0	0/24
FOM n1e-06, 11e-01	0.0002	0.0003	48.915	50.783	0/0	0/24

Table 1: Burgers 3D, 33 nodes per axis; provisional development comparison. Each method is evaluated on 8 distinct cases; all timed repetitions are retained. Errors are same-grid relative errors. NF/NS counts nonfinite/nonstationary cases; outliers are calls above 1.5 times the method median. Audited development data, same training membership. Errors use the initial-field norm and exclude time zero. Operator predictions interpolate the trained time knots to the common output times. Physical refinement covers only two cases; this panel supports same-grid comparisons only.



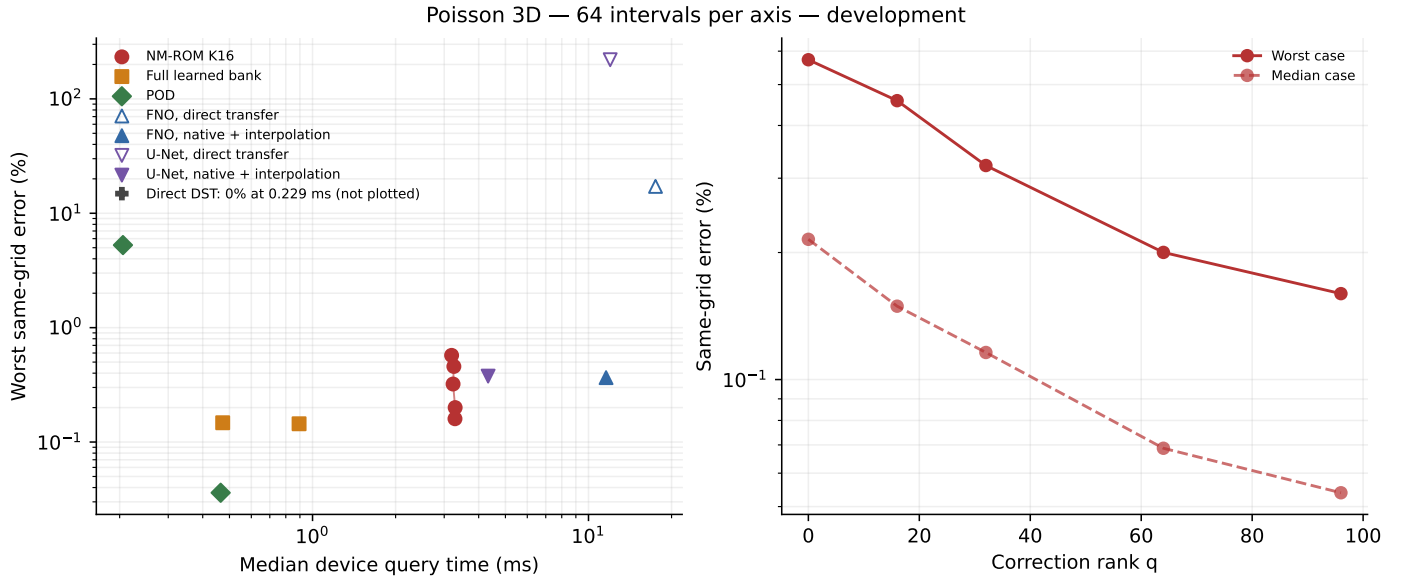
Method	Median (%)	Worst (%)	GPU ms	p95 ms	NF/NS	Outliers
NM-ROM K16, q0	1.0804	5.8427	18.392	36.053	0/2	6/48
NM-ROM K16, q8	1.0426	5.2535	18.294	23.042	0/0	0/48
NM-ROM K16, q32	0.8752	4.8852	19.608	21.957	0/0	0/48
NM-ROM K16, q64	0.5892	3.9930	20.528	24.009	0/0	0/48
NM-ROM K16, q96	0.3697	2.9628	24.581	37.254	0/1	3/48
Full bank, exact Galerkin	0.2563	1.3166	0.514	0.782	0/0	3/48
Full bank, exact weak	0.2542	1.3113	1.421	1.702	0/0	0/48
POD16	8.6633	25.1877	0.248	0.731	0/0	12/48
POD128	0.1414	1.2338	0.507	0.758	0/0	3/48
FNO, direct transfer	15.1234	17.6641	10.981	11.366	0/0	0/48
FNO, native + interpolation	0.7669	0.9891	7.140	7.855	0/0	0/48
U-Net, direct transfer	75.7115	92.6756	7.047	7.567	0/0	0/48
U-Net, native + interpolation	0.7512	0.9468	3.319	3.703	0/0	0/48
Direct DST	0.0000	0.0000	1.329	1.757	0/0	1/48

Table 2: Heat 3D, 64 intervals per axis; provisional development comparison. Each method is evaluated on 16 distinct cases; all timed repetitions are retained. Errors are same-grid relative errors. NF/NS counts nonfinite/nonstationary cases; outliers are calls above 1.5 times the method median. Audited development data. Errors use the current reference-field norm and exclude time zero. Some initial fits miss their stopping budget; evolved solves are stationary. Direct operator transfer and all quadrature certificates fail. Native-grid interpolation is charged separately.



Method	Median (%)	Worst (%)	GPU ms	p95 ms	NF/NS	Outliers
NM-ROM K16, q0	0.2149	0.5726	3.191	3.954	0/0	0/48
NM-ROM K16, q16	0.1492	0.4580	3.252	4.189	0/0	0/48
NM-ROM K16, q32	0.1159	0.3215	3.230	3.980	0/0	0/48
NM-ROM K16, q64	0.0687	0.2001	3.287	3.984	0/0	0/48
NM-ROM K16, q96	0.0539	0.1598	3.282	4.097	0/0	0/48
Full bank, weak solve	0.0520	0.1444	0.894	1.021	0/0	1/48
Full bank, Galerkin	0.0530	0.1474	0.473	0.726	0/0	3/48
POD16	1.8304	5.2702	0.206	0.673	0/0	8/48
POD128	0.0066	0.0360	0.465	0.764	0/0	5/48
FNO, direct transfer	16.0680	17.1866	17.493	18.200	0/0	0/48
FNO, native + interpolation	0.2852	0.3656	11.579	12.243	0/0	0/48
U-Net, direct transfer	168.0982	219.8928	11.985	12.627	0/0	0/48
U-Net, native + interpolation	0.2828	0.3769	4.328	4.887	0/0	0/48
Direct DST	0.0000	0.0000	0.229	0.581	0/0	10/48

Table 3: Poisson 3D, 64 intervals per axis; provisional development comparison. Each method is evaluated on 16 distinct cases; all timed repetitions are retained. Errors are same-grid relative errors. NF/NS counts nonfinite/nonstationary cases; outliers are calls above 1.5 times the method median. Development data. Static head-parameter transfers are included in this implementation’s timings; the next paired panel corrects their placement. Direct operator transfer and quadrature certification failed. All measurements below share one allocation.



Method	Median (%)	Worst (%)	GPU ms	p95 ms	NF/NS	Outliers
NM-ROM K16, q0	28.5319	50.4758	901.362	903.780	0/0	0/24
NM-ROM K16, q16	28.2539	49.9583	969.484	973.404	0/0	0/24
NM-ROM K16, q32	27.8899	49.2381	1092.342	1096.646	0/0	0/24
NM-ROM K16, q64	27.0931	47.6297	1588.925	1594.383	0/0	0/24
NM-ROM K16, q128	25.2664	45.3348	2226.148	2240.061	0/0	0/24
Full bank, Galerkin R512	13.4503	24.2696	49.405	49.601	0/0	0/24
POD16, weak solve	87.7996	91.9857	24.465	25.741	0/0	0/24
POD32, weak solve	78.1203	85.5263	27.381	28.384	0/0	0/24
POD48, weak solve	67.3545	76.1070	35.558	36.246	0/0	0/24
POD128, Galerkin	34.5481	47.3334	5.889	6.076	0/0	0/24
POD256, Galerkin	24.8410	36.0594	12.233	12.500	0/0	0/24
POD512, Galerkin	15.2837	27.7537	49.327	49.543	0/0	0/24
FOM, dt 0.001	0.0031	0.0051	25.763	26.674	0/0	0/24
FOM, dt 0.002	0.0130	0.0213	13.268	13.717	0/0	0/24
FOM, dt 0.004	0.0524	0.0866	7.056	7.507	0/0	0/24
FOM, dt 0.008	0.2125	0.3598	3.936	4.361	0/0	0/24
FNO, raw	1.3422	1.7664	9.554	10.080	0/0	0/24
FNO, projected	1.0974	1.5121	9.596	10.390	0/0	0/24
U-Net, raw	3.4540	4.6468	2.019	2.373	0/0	0/24
U-Net, projected	3.0414	4.2830	2.128	2.751	0/0	1/24

Table 4: Navier–Stokes 3D, 32 periodic points per axis; provisional development comparison. Each method is evaluated on 8 distinct cases; all timed repetitions are retained. Errors are same-grid relative errors. NF/NS counts nonfinite/nonstationary cases; outliers are calls above 1.5 times the method median. Audited negative development result on a fixed interacting three-dimensional flow family. Errors use the initial vector norm and exclude time zero. Projected operator variants charge the divergence-free projection; raw outputs are separate diagnostics. Full-bank/POD Galerkin controls are distinct reduced equations from the weak correction objective. Evolution stopping records have no saved latent history in this attempt.

